Evaluation Challenges of ML

- **Accuracy alone is misleading** in imbalanced datasets
- High accuracy can hide **critical failures**

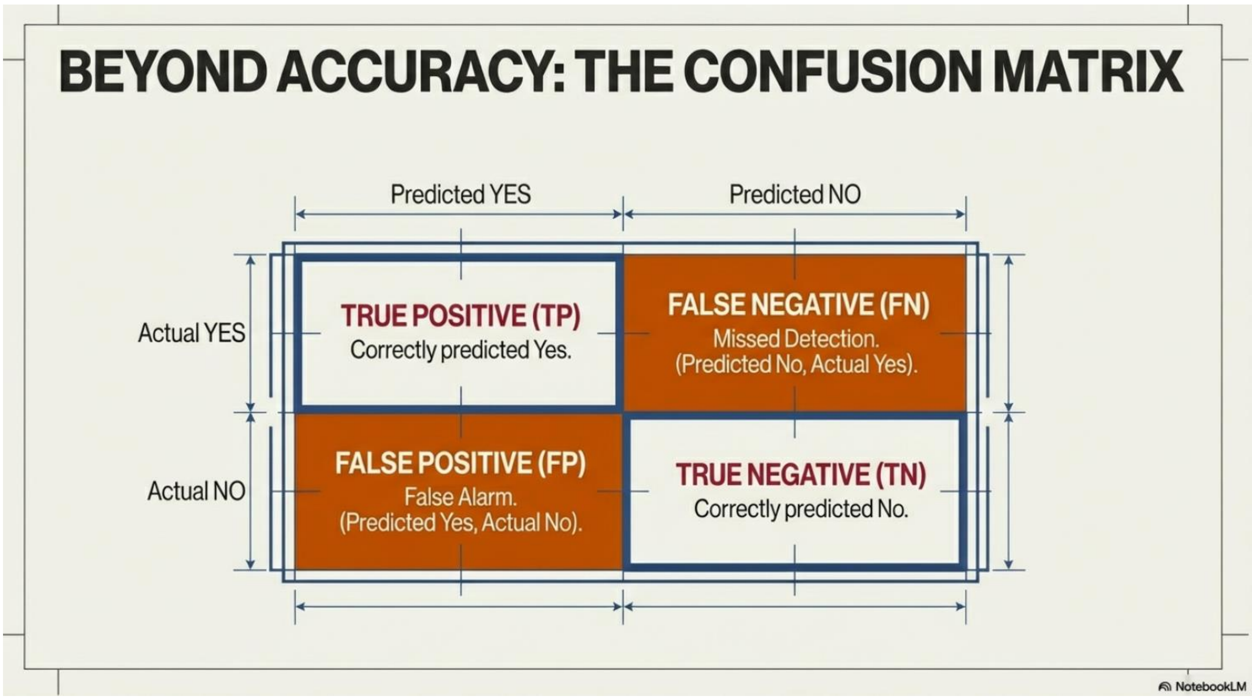
Examples:

- A cancer detection system with 99% accuracy may still fail if it misses positive cases.
- A credit card fraud model may classify all transactions as “non-fraud” and still show high accuracy.

- Confusion matrix is the **foundation** of evaluation
- Precision, Recall, and F1-score reveal **true model behavior**
- Always interpret metrics **together**, not in isolation

Confusion Matrix: The Core of All Metrics

A **confusion matrix** summarizes prediction results in a **2 × 2 table**.



Sklearn python library uses this exact order internally:

Actual \ Predicted	**Predicted Negative(0)**	**Predicted Positive(1)**
Actual Negative(0)	✔ **TN**	❌ **FP**
Actual Positive(1)	❌ **FN**	✔ **TP**

Meaning of Each Term (Real-World Mapping)

Term	Meaning	Cancer Example	Fraud Example
TP	Correct positive	Cancer detected	Fraud caught
FN	Missed positive	Cancer missed ❌	Fraud missed ❌
FP	False alarm	Healthy marked sick	Valid txn blocked
TN	Correct negative	Healthy confirmed	Valid txn allowed

👉 **Every evaluation metric is computed from these four values**

Why Accuracy Fails – Seen Through Confusion Matrix

Cancer Detection Example

Assume:

Actual \ Predicted	Cancer	No Cancer
Cancer	10 (TP)	90 (FN)
No Cancer	5 (FP)	9,895 (TN)

1. True Positive (TP) = 10

"Correctly caught cancer cases"

- These **10 people actually have cancer**
- The model **correctly predicted cancer**
- ✅ Good and desirable outcome

👉 *Out of all cancer patients, 10 were correctly identified as having cancer.*

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2. False Negative (FN) = 90

"Cancer cases that were missed"

- These **90 people actually have cancer**
- The model predicted **No Cancer**
- ❌ **Very dangerous mistake**

👉 *90 people have cancer, but the model said they are healthy.*

⚠ In real life, these patients may **not get treatment**.

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3. False Positive (FP) = 5

"False cancer alarms"

- These **5 people do NOT have cancer**
- The model predicted **Cancer**
- ❌ Unnecessary worry and tests

👉 *5 healthy people were wrongly told they might have cancer.*

⚠ Causes stress, extra tests, and cost.

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4. True Negative (TN) = 9,895

"Correctly identified healthy people"

- These **9,895 people are healthy**
- The model predicted **No Cancer**
- ✅ Correct and safe decision

👉 *Most healthy people were correctly told they are healthy.*

Final Remark: Out of 100 cancer patients, the model found only **10** and missed **90**, while correctly identifying most healthy people.

Why This Is Important

- **Accuracy looks high** because TN is huge (9,895)
- **But the model is unsafe** because FN is also huge (90)

- This is why **accuracy alone is misleading**

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Accuracy = $\frac{TP+TN}{Total} = \frac{10+9895}{10000} = 99.05\%$

Overall, the model is correct **99.05% of the time** (but this is misleading here).

Looks excellent; but,
❌ Misses **90% of cancer patients**

👉 Confusion matrix **exposes what accuracy hides**

Metrics Directly Derived from Confusion Matrix

Precision: "How Trustworthy Are Positive Predictions?"

$$\text{Precision} = \frac{TP}{TP + FP}$$

➡ Focuses on **prediction quality**

Example:

$$\frac{10}{10+5} = 0.67$$

10/15=0.6667 =0.67

Meaning:
When the model predicts Cancer, it is correct about 67% of the time.

Recall (Sensitivity): "Out of all actual positives, how many did we detect?"

$$\text{Recall} = \frac{TP}{TP + FN}$$

➡ Focuses on **coverage of actual positives**

Example:

$$\frac{10}{10 + 90} = 0.10$$

Meaning:
The model detects **only 10% of actual cancer patients**
👉 **90% are missed (very dangerous)**

F1-Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Balances both errors.

F1=2×(0.6667×0.10)/(0.6667+0.10)

F1=0.13334/0.7667 =0.1739

❌ F1-Score

0.1739=17.39%

Meaning:
Overall performance on the **cancer class is very poor**.

Specificity

$$Specificity = TN/(TN + FP)$$

Final Results

Metric	Value
Accuracy	**99.05%**
Precision	**66.67%**
Recall	**10%**
F1-score	**17.39%**

Key Insight

Although the accuracy is very high, the recall and F1-score are extremely low, indicating that the model fails to identify most cancer patients and is therefore unsuitable for medical diagnosis.

✓ Accuracy uses **all four**
✓ Precision uses **TP & FP**
✓ Recall uses **TP & FN**
✓ F1 combines **Precision & Recall**

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1. Why *Accuracy* Can Be Misleading

What is Accuracy?

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

It simply tells us:

"Out of all predictions, how many were correct?"

The Hidden Problem

Accuracy **treats all errors equally**, but in many real-world problems:

- **Some mistakes are far more costly than others**
- Data is often **highly imbalanced**

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2. Imbalanced Data: The Core Issue

Imbalanced dataset → One class appears much more frequently than the other.

Examples:

- Cancer detection → 95–99% patients are healthy
- Fraud detection → <1% transactions are fraud
- Spam detection → Most emails are not spam

In such cases, **predicting only the majority class** can give high accuracy while being **practically useless**.

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3. Example 1: Cancer Detection System

Scenario

- 10,000 patients
- 9,900 healthy

- 100 have cancer (1%)

Dumb Model

Predicts **“No Cancer” for everyone**

Actual / Predicted	**Cancer**	**No Cancer**
Cancer	0	100 (FN)
No Cancer	0	9,900 (TN)

Accuracy $= \frac{0+9900}{10000} = 99\%$

Looks amazing!

Reality Check

- **Missed ALL cancer patients**
- 100 people go untreated
- Model is **dangerous**

👉 Accuracy hides the real failure.

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4. Example 2: Credit Card Fraud Detection

Scenario

- 1,000,000 transactions
- 999,000 legitimate
- 1,000 fraud (0.1%)

Lazy Model

Predicts **“Not Fraud” always**

Accuracy $= \frac{999000}{1000000} = 99.9\%$

Business Impact

- Fraud loss = **huge**

- No fraud is ever detected
- Model provides **zero value**

👉 High accuracy ≠ good model

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5. Why We Need Better Metrics

Instead of asking:

"How many predictions were correct?"

We should ask:

- Did we **catch the important cases**?
- How many **false alarms** did we raise?
- How balanced is performance?

That's where **Precision, Recall, and F1-score** come in.

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6. Precision – "How Trustworthy Are Positive Predictions?"

$$\text{Precision} = \frac{TP}{TP + FP}$$

Meaning

"When the model predicts **Positive**, how often is it correct?"

Example (Fraud Detection)

- Model flags 200 transactions as fraud
- Only 50 are actually fraud

$$\text{Precision} = \frac{50}{200} = 0.25$$

👉 75% of alerts are false alarms
👉 Bad for customer experience

Important when:

- False positives are costly
- Example: spam filtering, fraud alerts

7. Recall – "How Much Did We Catch?"

$$\text{Recall} = \frac{TP}{TP + FN}$$

Meaning

"Out of all actual positives, how many did we detect?"

Example (Cancer Detection)

- 100 cancer patients
- Model detects 80

$$\text{Recall} = \frac{80}{100} = 0.8$$

👉 20 patients missed
👉 Dangerous but better than zero

Important when:

- Missing a positive case is costly
- Example: disease detection, fraud, terrorism

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8. Precision vs Recall Trade-off

Scenario	Priority
Cancer screening	**High Recall** (don't miss patients)
Spam filter	**High Precision** (don't block valid emails)
Fraud detection	**Recall first** (out of all actual frauds, how many did we catch), **then precision** (out of all transactions we flagged as fraud, how many were really fraud?)
Face recognition (security)	Precision critical

No single metric works everywhere.

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9. F1-Score – Balanced Evaluation

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Why F1?

- Penalizes extreme imbalance
- High only if **both precision and recall are high**
- Useful when:
    - Classes are imbalanced
    - Both false positives and false negatives matter

Example

- Precision = 0.6
- Recall = 0.4

$$F1 = 2 \times \frac{0.6 \times 0.4}{1.0} = 0.48$$

👉 Reflects mediocre performance honestly

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10. Summary Table

Metric	Answers	Best Used When
Accuracy	Overall correctness	Balanced datasets
Precision	“Can I trust positive predictions?”	False alarms costly
Recall	“Did I catch important cases?”	Misses are costly
F1-score	Balanced performance	Imbalanced data

✅ What is Balanced Data?

Balanced data means:

The number of samples in each class is **approximately equal**.

📌 Example 1 (Balanced Dataset)

Suppose we have 100 emails:

- Spam = 50
- Not Spam = 50

Here both classes are equal → ✅ **Balanced Dataset**

📌 Example 2 (Imbalanced Dataset)

Suppose we have 1000 transactions:

- Fraud = 20
- Genuine = 980

Here one class is much smaller → ❌ **Imbalanced Dataset**

If **Minority class < 20% of total**
It is generally considered **imbalanced**.

🎯 Why Balanced Data Matters?

Because evaluation metrics behave differently.

🔎 In Balanced Data

Example:

Class	Count
Positive	50
Negative	50

If model predicts 90 correctly:

Accuracy = 90%

👉 Accuracy is meaningful here because both classes are equally represented.

⚠ In Imbalanced Data

Example:

Class	Count
Fraud	20
Genuine	980

If model predicts ALL as Genuine:

Accuracy = 980 / 1000 = **98%**

But it detects **0 frauds** 😨

👉 Accuracy becomes misleading.

🎯 ACCURACY – When Overall Correctness Matters

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

🔷 When Important?

✅ Condition:

- Classes are balanced
- FP and FN have similar cost

📌 Example 1: Cats vs Dogs Image Classification

🎯 Metric: **Accuracy**

🔍 What error is dangerous?
Neither FP nor FN is more dangerous.

📌 Reason:
Misclassifying a cat as dog is not critical.
Both types of errors are equally acceptable.

So overall correctness matters most.

📌 Example 2: Handwritten Digit Recognition

🎯 Metric: **Accuracy**

🔍 What error is dangerous?
All errors are equally bad.

📌 Reason:
If digit 3 is classified as 8, it's just wrong.
No special danger.

So we want maximum overall correct predictions.

📌 **Example 3: Balanced Sentiment Analysis**

🎯 Metric: **Accuracy**

🔍 What error is dangerous?
Positive and negative misclassifications have similar impact.

📌 Reason:
No class is more important.

🎯 PRECISION

What Precision Controls

$$Precision = \frac{TP}{TP + FP}$$

👉 **Precision mainly controls:**

- **False Positives (FP)**

High Precision means:
✓ When model predicts "Positive", it is usually correct
❌ Very few false alarms

📌 PRECISION IS IMPORTANT WHEN:

False Positives are costly, dangerous, embarrassing, or financially harmful.

📧 1️⃣ Spam Detection

Problem:

Classify email as Spam (Positive) or Not Spam (Negative)

Dangerous Error:

False Positive
(Important email wrongly marked as spam)

Why Precision Matters?

If precision is low:

- Many genuine emails go to spam
- User misses important messages
- Trust in system decreases

So we want:
When system says “Spam” → it should almost always be spam.

👉 High Precision required.

💳 2️⃣ Credit Card Blocking

Problem:

Detect fraudulent transaction

Dangerous Error:

False Positive
(Genuine transaction blocked)

Why Precision Matters?

If precision is low:

- Customer’s card gets blocked unnecessarily
- Customer frustration
- Business reputation affected

So:
When system says “Fraud” → it must be truly fraud.

👉 High Precision important.

🧑⚖️ 3️⃣ Court Verdict Prediction

Problem:

Predict whether accused is guilty (Positive)

Dangerous Error:

False Positive
(Innocent predicted guilty)

Why Precision Matters?

False conviction is morally and legally unacceptable.

When model says “Guilty” → must be very confident.

👉 High Precision required.

🏦 4️⃣ Loan Approval / Default Prediction

Problem:

Predict customer will default (Positive)

Dangerous Error:

False Positive
(Customer predicted risky but actually safe)

Why Precision Matters?

If bank wrongly labels safe customer as risky:

- Loan rejected unfairly
- Business loss
- Customer dissatisfaction

So when model predicts “High Risk”, it must be correct.

👉 High Precision required.

🔐 5️⃣ Biometric Authentication

Problem:

Grant system access

Dangerous Error:

False Positive
(Unauthorized person granted access)

Why Precision Matters?

If system says "Authorized" incorrectly:

- Security breach
- Data theft
- Serious consequences

So:
When model says "Access Granted" → must be correct.

👉 High Precision required.

🧪 6️⃣ Medical Confirmatory Test (Second Test)

Problem:

Confirm disease

Dangerous Error:

False Positive
(Healthy person diagnosed sick)

Why Precision Matters?

If precision is low:

- Healthy person undergoes unnecessary treatment
- Emotional stress
- Financial burden

So:
When test says "Disease" → it should almost always be correct.

👉 High Precision important.

📊 Precision Logic in Simple Terms

Ask this:

👉 “If system says Positive, can I trust it?”

If answer must be YES → Precision is important.

🎯 RECALL – When Missing Positive is Dangerous

$$Recall = \frac{TP}{TP + FN}$$

📌 Example 1: Cancer Detection

🎯 Metric: **Recall**

🔍 Dangerous Error: **False Negative**

📌 Reason:
If cancer patient is predicted healthy → treatment delayed → life risk.

So FN must be minimized → Recall must be high.

📌 Example 2: Fraud Detection

🎯 Metric: **Recall**

🔍 Dangerous Error: False Negative

📌 Reason:
If fraud is not detected → financial loss.

Better to block extra transactions than miss fraud.

📌 Example 3: Fire Detection System

🎯 Metric: **Recall**

🔍 Dangerous Error: Missing fire

📌 Reason:
False alarm is irritating.
Missing fire is catastrophic.

📌 **Example 4: Terrorist Detection**

🎯 Metric: **Recall**

🔍 Dangerous Error: Missing threat

📌 Reason:
False alarm manageable.
Missing terrorist → security disaster.

🎯 **SPECIFICITY – When Correctly Identifying Negatives is Important**

$$Specificity = \frac{TN}{TN + FP}$$

📌 **Example 1: Confirmatory Medical Test**

🎯 Metric: **Specificity**

🔍 Dangerous Error: False Positive

📌 Reason:
Wrongly diagnosing healthy person causes stress and unnecessary treatment.

So FP must be minimized.

📌 **Example 2: Factory Defect Testing**

🎯 Metric: **Specificity**

🔍 Dangerous Error: Rejecting good products

📌 Reason:
Rejecting non-defective items causes financial loss.

📌 **Example 3: Drug Trial Approval**

🎯 Metric: **Specificity**

🔍 Dangerous Error: Approving ineffective drug

📌 Reason:
Wrong approval can harm patients.

🎯 F1 SCORE – When Both Precision & Recall Matter

$$F1 = \frac{2TP}{2TP + FP + FN}$$

📌 Example 1: Credit Card Fraud Detection

🎯 Metric: **F1 Score**

🔍 Dangerous Errors:

- Missing fraud (FN)
- Blocking genuine transaction (FP)

📌 Reason:
We must balance both.

High recall + high precision required.

📌 Example 2: Rare Disease Detection

🎯 Metric: **F1 Score**

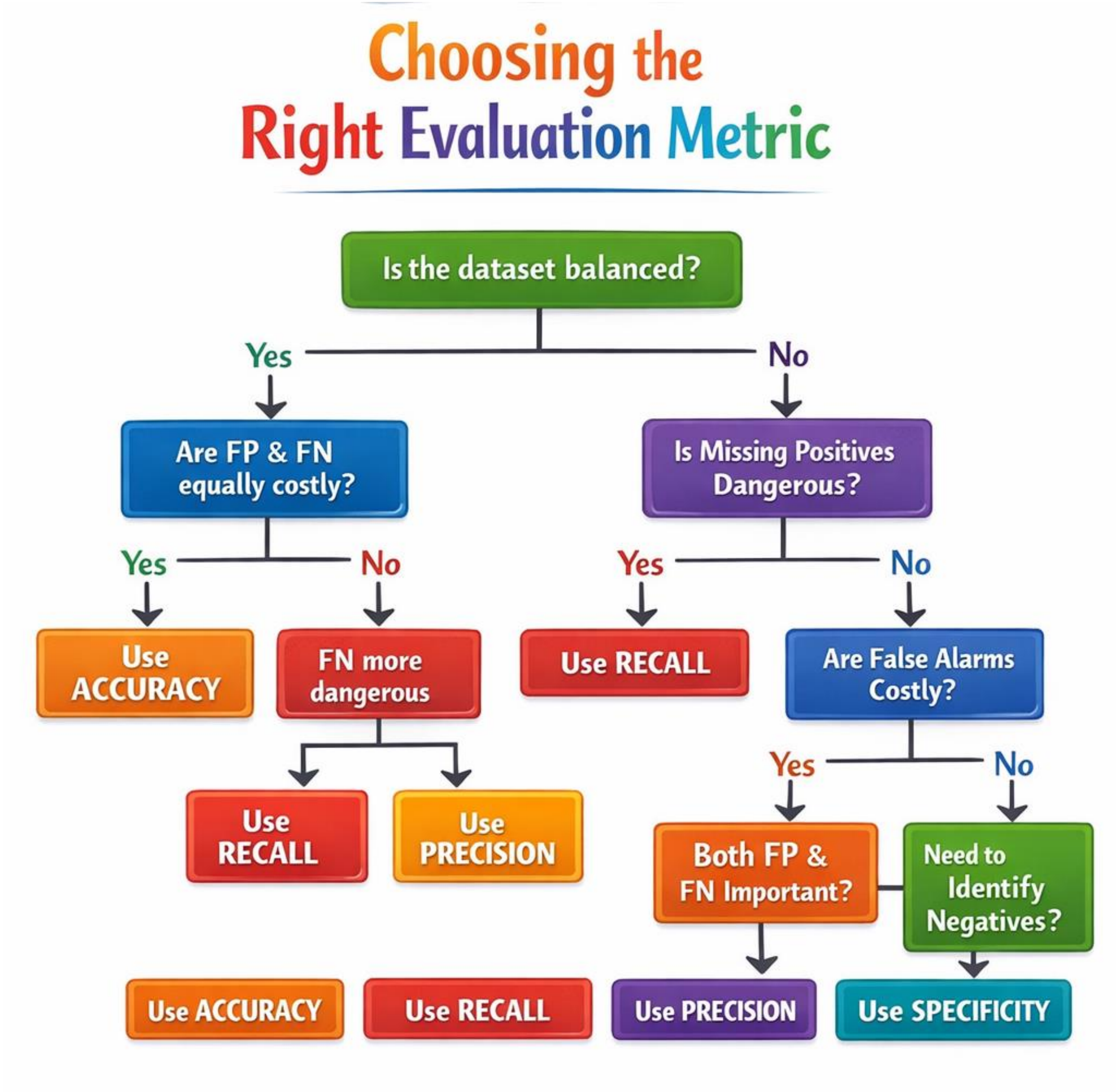
🔍 Dangerous Errors:

- Missing patient (FN)
- Wrongly alarming healthy person (FP)

📌 Reason:
Disease is rare → imbalanced data
Accuracy becomes misleading
Need balance → F1

📊 Quick Classification Table

Problem Type	**Most Important Metric**	**Why**
Balanced image classification	Accuracy	Overall correctness
Spam filter	Precision	Avoid false alarms
Cancer detection	Recall	Avoid missing cases
Confirmatory medical test	Specificity	Avoid wrong diagnosis
Fraud detection	F1	Balance precision & recall
Fire alarm system	Recall	Missing fire is dangerous
Loan approval	Precision	Avoid risky approvals
Rare disease screening	Recall	Catch all cases



PROBLEM 1: Spam Detection (Balanced Data)

An email service provider is developing a spam detection system.

A dataset of 20 emails is used to test the model.

- 1 represents Spam
- 0 represents Not Spam

The actual classification of the emails is:

1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0

A machine learning model (Model A) produces the following predictions:

1, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 1, 0, 0, 0, 1, 0, 1, 0

Questions:

(a) Construct the Confusion Matrix for Model A.

(b) Calculate the following performance metrics:

- Accuracy
- Precision
- Recall (Sensitivity)
- Specificity
- F1-score

(c) Interpret the results of the model in the context of spam detection.

(d) Which metric is most important in spam detection? Justify your answer.

📌 Dataset (20 Emails)

Actual (1 = Spam, 0 = Not Spam)

Actual:
[1,0,1,0,1,0,1,0,1,0, 1,0,1,0,1,0,1,0,1,0]

🔷 Model A Predictions

Predicted A:
[1,0,1,0,1,0,0,0,1,0, 1,1,1,0,0,0,1,0,1,0]

Method to calculate TP, TN, FN, FP:

- TP → Count how many times both lists have 1 in same position
- TN → Count how many times both lists have 0 in same position
- FN → Actual 1 but Predicted 0 (Missed Detection)
- FP → Actual 0 but Predicted 1 (False Alarm)

Positive(1), Negative(0)	**Actual**	**Predicted**
True Positive(TP)	1	1
True Negative(TN)	0	0
False Positive(FP)	0	1
False Negative(FN)	1	0

Count Values

- **TP (Spam correctly detected)** = 8
- **FN (Spam missed)** = 2
- **FP (Not spam marked as spam)** = 1
- **TN (Correctly identified not spam)** = 9

Check total:
8 + 2 + 1 + 9 = 20 ✓

📊 Confusion Matrix

	Predicted Spam	Predicted Not Spam
Actual Spam	8 (TP)	2 (FN)
Actual Not Spam	1 (FP)	9 (TN)

(b) Performance Metrics

1️⃣ Accuracy

$$Accuracy = \frac{TP + TN}{Total}$$
$$= \frac{8+9}{20}$$
$$= \frac{17}{20} = 0.85$$

✅ Accuracy = 85%

2️⃣ Precision

$$Precision = \frac{TP}{TP + FP}$$
$$= \frac{8}{8+1}$$
$$= \frac{8}{9} = 0.889$$

✅ Precision ≈ 88.9%

3 Recall (Sensitivity)

$$Recall = \frac{TP}{TP + FN}$$
$$= \frac{8}{8 + 2}$$
$$= \frac{8}{10} = 0.8$$

✅ **Recall = 80%**

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4 Specificity

$$Specificity = \frac{TN}{TN + FP}$$
$$= \frac{9}{9 + 1}$$
$$= \frac{9}{10} = 0.9$$

✅ **Specificity = 90%**

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5 F1-Score

$$F1 = \frac{2PR}{P + R}$$
$$= \frac{2 \times 0.889 \times 0.8}{0.889 + 0.8}$$
$$= \frac{1.4224}{1.689}$$
$$\approx 0.842$$

✅ **F1-score ≈ 84.2%**

(c) Interpretation in Spam Detection Context

- The model correctly detects **80% of spam emails**.
- It rarely mislabels genuine emails as spam (only 1 false alarm).
- Precision is high (88.9%), meaning when it flags spam, it is usually correct.
- Specificity is also strong (90%), so genuine emails are mostly safe.

Overall:

✓ Good spam detection
✓ Few false alarms
✓ Still misses 2 spam emails

(d) Most Important Metric in Spam Detection?

✅ Most Important: Precision

Why?

In spam detection:

- False Positive = Important genuine email goes to spam folder ❌
- This can cause serious inconvenience (missing job emails, bank alerts, etc.)

High Precision ensures:

👉 If the system says "Spam", it is very likely spam.
👉 Important emails are not wrongly blocked.

🎯 Final Conclusion

Model A performs well:

- Accuracy = 85%
- Precision is high (88.9%)
- Recall is decent (80%)
- F1-score balanced (84%)

It is a reasonably good spam detection model, but improving Recall slightly would make it stronger.

🔷 Model B Predictions

Predicted B:
[1,0,1,0,1,0,1,0,1,0, 1,0,1,0,1,0,1,0,1,0]

Confusion Matrix (Model B)

	Pred 1	**Pred 0**
Actual 1	TP = 10	FN = 0
Actual 0	FP = 0	TN = 10

All spam caught.
No normal email wrongly blocked.

All metrics = 1.

👉 Ideal model.

Metrics (Model B)

Accuracy = 20/20 = **1.00**

Precision = 1.00
Recall = 1.00
Specificity = 1.00
F1 = 1.00

✅ Best Model?

Model B (Perfect classifier)

✅ PROBLEM 2: Cancer Detection (Imbalanced Data)

A cancer detection system was tested on 1000 patients, out of which 100 were actual cancer patients and 900 were healthy. The model correctly identified 20 cancer patients and 890 healthy patients. It misclassified 80 cancer patients as healthy and 10 healthy patients as cancer patients.

Construct the confusion matrix and calculate Accuracy, Precision, Recall, and Specificity.

Comment on whether the model is suitable for cancer detection.

Total Patients = 1000
Cancer Patients = 100
Healthy = 900

🔷 Model A (High Accuracy but Poor Recall)

Confusion Matrix:

	Cancer	**No Cancer**
Cancer	TP = 20	FN = 80
Healthy	FP = 10	TN = 890

Metrics

Accuracy = (20+890)/1000 = **0.91**

Precision = 20/(20+10) = **0.67**

Recall = 20/(20+80) = **0.20**

Specificity = 890/(890+10) = **0.99**

F1 = **0.31**

⚠ Accuracy = 91%

Looks very good.

But what does it hide?

🚨 Recall = 0.20 (20%)

Out of 100 cancer patients:
Only 20 detected.
80 missed.

👉 80 patients go untreated.

This is extremely dangerous.

Precision = 0.67

When model says cancer, it is correct 67% of time.

Specificity = 0.99

Almost all healthy people correctly identified.

👉 Model is excellent at identifying healthy patients.

F1 = 0.31

Poor balance.

🔎 Real Meaning of Model A

It is basically predicting “Healthy” most of the time.

Because healthy people are majority (900).

That’s why accuracy is high.

But it fails in actual purpose: detecting cancer.

🔷 Model B (Better Recall)

	Cancer	**No Cancer**
Cancer	TP = 80	FN = 20
Healthy	FP = 90	TN = 810

Metrics

Accuracy = (80+810)/1000 = **0.89**

Precision = 80/(80+90) = **0.47**

Recall = 80/(80+20) = **0.80**

Specificity = 810/(810+90) = **0.90**

F1 = **0.60**

Accuracy = 89%

Lower than Model A.

But accuracy is misleading here.

🚀 Recall = 0.80 (80%)

Out of 100 cancer patients:
80 detected.
Only 20 missed.

Much safer.

Precision = 0.47

Many false alarms (90 healthy predicted cancer).

But in medicine, false alarm is better than missing cancer.

F1 = 0.60

Much better balance than Model A.

✅ **Best Model?**

Even though Model A has higher accuracy (91%),
Model B is better for cancer detection because:

✓ High Recall (0.80)
✓ Higher F1 score

👉 In medical diagnosis → **Recall is more important than Accuracy**

✅ **PROBLEM 3: Credit Card Fraud**

A bank used a machine learning model to detect fraudulent transactions among 10,000 transactions. Out of 200 actual frauds, 150 were correctly identified and 50 were missed. Out of 9,800 genuine transactions, 300 were wrongly classified as fraud.

(a) Construct the confusion matrix
(b) Calculate Accuracy, Precision, Recall, Specificity, and F1-score
(c) Comment whether this model is suitable for fraud detection.

Total = 10,000 Transactions
Fraud = 200
Genuine = 9800

🔷 **Model A**

	Fraud	**Genuine**
Fraud	TP = 150	FN = 50
Genuine	FP = 300	TN = 9500

Metrics

Accuracy = (150+9500)/10000 = **0.965**

Precision = 150/(150+300) = **0.33**

Recall = 150/(150+50) = **0.75**

Specificity = 9500/(9500+300) = **0.97**

F1 = **0.46**

Accuracy = 96.5%

Looks very high.

But data is imbalanced.

Recall = 0.75

Catches 75% fraud.

50 fraud cases missed.

Loss occurs.

⚠ **Precision = 0.33**

Out of 450 flagged transactions,
Only 150 are truly fraud.

300 genuine customers wrongly blocked.

👉 Very bad customer experience.

F1 = 0.46

Poor balance.

🔷 **Model B**

	Fraud	**Genuine**
Fraud	TP = 130	FN = 70
Genuine	FP = 50	TN = 9750

Metrics

Accuracy = (130+9750)/10000 = **0.988**

Precision = 130/(130+50) = **0.72**

Recall = 130/(130+70) = **0.65**

Specificity = 9750/(9750+50) = **0.995**

F1 = **0.68**

Accuracy = 98.8%

Very high.

Precision = 0.72

Only 50 genuine transactions wrongly blocked.

Much better than Model A.

Recall = 0.65

Misses slightly more fraud than Model A.

F1 = 0.68

Better balance overall.

✅ **Best Model?**

Model A:
✓ Higher recall
❌ Terrible precision

Model B:
✓ Good precision
✓ Good specificity
✓ Better F1
⚠ Slightly lower recall

👉 In fraud detection, blocking too many genuine customers is bad.

So balance is required.

Model B is better overall.

👉 In fraud detection → balance between Precision & Recall is important.

Conclusion:

Accuracy alone is NOT enough.

You must always ask:

1. Is data balanced?
2. Which error is more dangerous?
3. Do I need balance?

Solve the following problems:

Question 1:

A spam detection model was tested on 20 emails. Out of 10 actual spam emails, 8 were correctly detected. Out of 10 legitimate emails, 1 was wrongly classified as spam.

(a) Construct the confusion matrix
(b) Calculate evaluation metrics
(c) Which metric is most important in spam detection? Justify.

Total emails = 20
Actual Spam = 10
Actual Legitimate = 10

Given:

- 8 spam correctly detected → **True Positive (TP) = 8**
- 10 − 8 = 2 spam missed → **False Negative (FN) = 2**
- 1 legitimate wrongly classified as spam → **False Positive (FP) = 1**
- Remaining legitimate correctly classified → 10 − 1 = **True Negative (TN) = 9**

(a) Confusion Matrix

	Predicted Spam	**Predicted Legitimate**
Actual Spam	TP = 8	FN = 2
Actual Legitimate	FP = 1	TN = 9

(b) Evaluation Metrics

1 **Accuracy**

$$Accuracy = \frac{TP + TN}{Total}$$
$$= \frac{8 + 9}{20} = \frac{17}{20} = 0.85$$

✅ **Accuracy = 85%**

2️⃣ Precision

$$Precision = \frac{TP}{TP + FP}$$
$$= \frac{8}{8 + 1} = \frac{8}{9} = 0.889$$

✅ **Precision ≈ 88.9%**

Meaning: When the model says “Spam”, it is correct 88.9% of the time.

3️⃣ Recall (Sensitivity)

$$Recall = \frac{TP}{TP + FN}$$
$$= \frac{8}{8 + 2} = \frac{8}{10} = 0.8$$

✅ **Recall = 80%**

Meaning: It catches 80% of actual spam emails.

4️⃣ F1-Score

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$
$$= \frac{2 \times 0.889 \times 0.8}{0.889 + 0.8}$$
$$= \frac{1.4224}{1.689} \approx 0.842$$

✅ **F1 ≈ 84.2%**

(c) Most Important Metric in Spam Detection?

🎯 **Precision is usually most important.**

Why?

- If a legitimate email is marked as spam (False Positive),
    - Important emails may be lost.
    - User trust decreases.

In this example:

- Only 1 legitimate email was wrongly blocked.
- Precision is high (88.9%) → Good.

Summary Table

Metric	Value
Accuracy	85%
Precision	88.9%
Recall	80%
F1-score	84.2%

✅ Final Answer

Although the model has 85% accuracy, the most important metric in spam detection is **Precision**, because incorrectly classifying legitimate emails as spam is more harmful than missing some spam emails. High precision ensures genuine emails are not wrongly blocked.

Question 2:

A disease screening test was conducted on **10,000 people**.

Only **100 people actually have the disease**.

A model predicted:

- 95 diseased people correctly.
- 5 diseased people as healthy.
- 500 healthy people as diseased.
- 9,400 healthy people correctly.

The model achieved **94.95% accuracy**.

Questions:

1. Construct the confusion matrix.
2. Calculate Precision, Recall, and F1-score.
3. Despite high accuracy, explain why this model may still not be ideal.
4. Which metric should be prioritized in this problem?

This is a **classic imbalanced dataset problem** (very important for exams + ML interviews).

Total people = **10,000**
Actual Diseased = **100**
Actual Healthy = **9,900**

Model predictions:

- 95 diseased correctly → **True Positive (TP) = 95**
- 5 diseased predicted healthy → **False Negative (FN) = 5**
- 500 healthy predicted diseased → **False Positive (FP) = 500**
- 9,400 healthy correctly predicted → **True Negative (TN) = 9400**

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1 Confusion Matrix

	Predicted Diseased	**Predicted Healthy**
Actual Diseased	TP = 95	FN = 5
Actual Healthy	FP = 500	TN = 9400

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2 Accuracy (Given = 94.95%)

$$Accuracy = \frac{TP + TN}{Total}$$
$$= \frac{95 + 9400}{10000}$$
$$= \frac{9495}{10000}$$
$$= 94.95\%$$

✓ Correct.

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3 Precision

$$Precision = \frac{TP}{TP + FP}$$
$$= \frac{95}{95 + 500}$$
$$= \frac{95}{595}$$
$$\approx 0.1597$$
$$Precision \approx 15.97\%$$

⚠ Very low!

Meaning: When model says "diseased", it is correct only 16% of the time.

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4 Recall (Sensitivity)

$$Recall = \frac{TP}{TP + FN}$$
$$= \frac{95}{95 + 5}$$
$$= \frac{95}{100}$$
$$= 0.95$$
$$Recall = 95\%$$

✓ Very high!

Meaning: It catches 95% of actual diseased patients.

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5 F1-Score

$$
\begin{aligned}
F1 &= \frac{2 \times Precision \times Recall}{Precision + Recall} \\
&= \frac{2 \times 0.1597 \times 0.95}{0.1597 + 0.95} \\
&= \frac{0.303}{1.1097} \\
&\approx 0.273 \\
F1 &\approx 27.3\%
\end{aligned}
$$

⚠ Very low overall balance.

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6 Why High Accuracy is Misleading?

Because the dataset is **highly imbalanced**:

- Diseased = 100
- Healthy = 9,900

Even if the model predicts **everyone as healthy**, accuracy would be:

$$\frac{9900}{10000} = 99\%$$

But it would detect **zero diseased patients**.

In our case:

- Accuracy = 94.95% (looks great!)
- But Precision = 16% (terrible)

The model generates **too many false alarms (500 FP)**.

That means:

- 500 healthy people will panic.
- Unnecessary tests, cost, stress.

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7 Which Metric Should Be Prioritized?

In **disease screening problems**, we usually prioritize:

✅ Recall (Sensitivity)

Because:

- Missing a diseased patient (FN) is very dangerous.
- Early detection is critical.

So Recall is most important.

However:

- If false positives are too high (like here),
- Precision also becomes important.

Best Practical Choice:

✔ High Recall
✔ Then improve Precision
✔ Or use **F1-score** for balance

🎯 Final Interpretation

Metric	Value	Interpretation
Accuracy	94.95%	Misleading
Precision	15.97%	Very poor
Recall	95%	Excellent
F1-score	27.3%	Poor balance

🔥 Final Conclusion

Although the model achieves 94.95% accuracy, it performs poorly in terms of precision due to a large number of false positives. Since disease detection is a critical application where missing a patient can be dangerous, **Recall should be prioritized**. Accuracy is not a reliable metric in imbalanced datasets.